

GROUP THEORY

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Binary operation: Let A be a non-empty set. ⁴⁷

The mapping defined from $A \times A$ to A is called a Binary composition on A . It is denoted by $*$ (star), $\circ$ (circle), $\cdot$ (dot) etc.

i.e., $a, b \in A \Rightarrow a * b \in A$ then $*$ is a binary operation on A .

Group: Let G be a non-empty set and $*$ is a binary composition on A . The algebraic structure $(G, *)$ is said to be a group if it satisfies the following axioms:

G_1 : closure axiom: $a, b \in G \Rightarrow a * b \in G$

G_2 : Associative axiom:

$$a, b, c \in G \Rightarrow (a * b) * c = a * (b * c)$$

G_3 : Identity axiom:

For $a \in G \exists e \in G$ such that $a * e = e * a = a$
Here e is the identity element.

G_4 : Inverse axiom:

For $a \in G \exists b \in G$ such that $a * b = b * a = e$
Here $b = a^{-1}$ is the inverse element of a .

$$\text{i.e., } a * a^{-1} = a^{-1} * a = e$$

Abelian group: (commutative group)

A group $(G, *)$ is said to be an Abelian group if $a, b \in G$ then $a * b = b * a$.

G_5 : Abelian (commutative) axiom:

$$a, b \in G \Rightarrow a * b = b * a$$

NOTE : ① An algebraic structure $(G, *)$ is said to be a Semi-Group if it satisfies 48 G_1, G_2 .

② An algebraic structure $(G, *)$ is said to be a Monoid, if it satisfies G_1, G_2, G_3 .

Examples :

- ① $(N, +)$ is a semi-group but not a group because Identity does not exist.
- ② $(N, \times)$ is a monoid but not a group because $a \in N \Rightarrow a^{-1} = \frac{1}{a} \notin N$. Inverse axiom not satisfied.
- ③ Let G^+ be the set of positive rationals. The binary composition $*$ is defined by $a * b = \frac{ab}{5}$ $\forall a, b \in G^+$ Then p.t $(G^+, *)$ is an abelian group.
- ④ prove that the set of all real numbers other than 1 is an abelian group w.r.t the binary composition $*$ defined by $a * b = a + b - ab$.
- ⑤ p.t the set $G = \left\{ \begin{pmatrix} x & x \\ 0 & 0 \end{pmatrix} \mid x \in \mathbb{R}, x \neq 0 \right\}$ is an abelian group w.r.t multiplication.
- ⑥ p.t the set $G = \left\{ \begin{pmatrix} x & x \\ x & x \end{pmatrix} \mid x \in \mathbb{R}, x \neq 0 \right\}$ is an abelian group w.r.t matrix multiplication.

Order of group: The no of different elements of the group is the order of group. It is denoted by $o(G)$.

Properties of Groups:

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(2)

Theorem (1): Cancellation Laws :

If $(G, *)$ is a group and $a, b, c \in G$ then

- (i) $a * b = a * c \Rightarrow b = c$ (Left cancellation Law)
- (ii) $b * a = c * a \Rightarrow b = c$ (Right cancellation Law).

Proof: Given $(G, *)$ is a group and $a, b, c \in G$
since G is a group and $a \in G \Rightarrow a^{-1} \in G$

(i) consider $a * b = a * c$

pre operating a^{-1} on both sides,

$$a^{-1} * (a * b) = a^{-1} * (a * c)$$

$$\Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c \quad \text{Associative Law}$$

$$\Rightarrow e * b = e * c \quad \text{Inverse Law}$$

$$\Rightarrow b = c \quad \text{Identity Law.}$$

(ii) consider $b * a = c * a$

post operating a^{-1} on both sides

$$(b * a) * a^{-1} = (c * a) * a^{-1}$$

$$\Rightarrow b * (a * a^{-1}) = c * (a * a^{-1}) \quad \text{Associative Law}$$

$$\Rightarrow b * e = c * e \quad \text{Inverse Law}$$

$$\Rightarrow b = c \quad \text{Identity Law.}$$

$\therefore$ In a group, Cancellation Laws holds good.

Theorem (2): In a group, the identity element is unique.

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Proof: Let $(G, *)$ be a group and e be the identity element.

If possible, let e_1 and e_2 be two identity elements of G .

Since e_1 is identity, $e_1 * e_2 = e_2 * e_1 = e_1$

Since e_2 is identity, $e_2 * e_1 = e_1 * e_2 = e_2$

$\therefore$ In both the cases, we get $e_1 = e_2$

$\therefore$ Identity element is unique.

Theorem (3): In a group, the inverse of any element is unique.

Proof: Let $(G, *)$ be a group and e be the identity element. Let $a \in G$ be any element.

If possible, let b and c be two inverse of a .

i.e., $a * b = b * a = e$ and $a * c = c * a = e$

Now $b * (a * c) = (b * a) * c$

$$b * e = e * c$$

$$b = c$$

$\therefore$ Hence the inverse element is unique.

Subgroup: A nonempty subset H of a group G is said to be a subgroup if H itself is a group.

Ex: (i) The set $H = \{1, -1\}$ is a subgroup of the multiplicative group $G = \{1, -1, i, -i\}$.

(ii) The set of all even integers is a subgroup of the additive group of integers.

(iii) $G = \{1, -1, i, -i\}$ is a multiplicative group.

The subgroups of G are $H_1 = \{1\}$, $H_2 = \{1, -1\}$, $H_3 = \{1, -1, i, -i\} = G$. Here H_1 and H_3 are called Improper subgroups of G and H_2 is called proper subgroup.

(iv) If G is a group then $H = \{e\}$ and $H = G$ are two improper subgroups.

NOTE :-

① If H is a subgroup of G then the identity element in H and G are same.

② If H is a subgroup of a group G then the inverse of any element in H is same as the inverse of that element in G .

Theorem ①: The necessary and sufficient conditions for a non-empty subset H of a group G to be a subgroup are

$$(i) a, b \in H \Rightarrow a \cdot b \in H$$

$$(ii) a \in H \Rightarrow a^{-1} \in H.$$

Sol:- Case ①: Suppose H is a subgroup of G 46

By def H satisfies all properties of group G .

$\therefore$ By closure Law, $a, b \in H \Rightarrow a \cdot b \in H$

Let $a \in H$. By Inverse Law, $a^{-1} \in H$.

Case ②: Suppose H is a subset of G and

(i) $a, b \in H \Rightarrow a \cdot b \in H$ (ii) $a \in H \Rightarrow a^{-1} \in H$.

To prove H is a subgroup of G .

Closure Law: Given $a, b \in H \Rightarrow a \cdot b \in H$

Associative Law: Since $H \subset G$, all the elements of H satisfy associative law.

Identity Law: Given $a \in H, a^{-1} \in H \Rightarrow a \cdot a^{-1} \in H$
 $\Rightarrow e \in H$

$\therefore H$ contains an identity element.

Inverse Law: Given $a \in H \Rightarrow a^{-1} \in H$

$\therefore H$ is a subgroup of G .

Theorem ②: The necessary and sufficient condition for a non-empty subset H of a group G to be a subgroup is that $a, b \in H \Rightarrow a \cdot b^{-1} \in H$.

Proof:- Case ①: Suppose H is a subgroup of G

To prove $a, b \in H \Rightarrow a \cdot b^{-1} \in H$.

Since H is a subgroup, $b \in H \Rightarrow b^{-1} \in H$.

$\therefore a \in H, b^{-1} \in H \Rightarrow a \cdot b^{-1} \in H$

Case ②:- Suppose H is a non-empty subset of G

and $a, b \in H \Rightarrow a \cdot b^{-1} \in H$.

To prove H is a subgroup of G .

(i) Identity Law: put $b=a$.

$$a, a \in H \Rightarrow a \cdot a^{-1} \in H \Rightarrow e \in H$$

$\Rightarrow H$ contains identity.

(ii) Inverse Law: put $a=e$

$$e, b \in H \Rightarrow e \cdot b^{-1} \in H$$

$$\Rightarrow b^{-1} \in H$$

$\therefore H$ contains inverse element.

(iii) closure Law: put $b=b^{-1}$

$$a, b^{-1} \in H \Rightarrow a \cdot (b^{-1})^{-1} \in H \Rightarrow a \cdot b \in H$$

(iv) since $H \subset G$, all the elements of H satisfied Associative law.

Hence H is a subgroup of G .

NOTE :- ① In a group G , the intersection of two subgroups is a subgroup.

② In a group G , the union of two subgroups is a subgroup of G iff one subgroup is contained in another.

Ex: $(\mathbb{Z}, +)$ is a group

Let $H_1 = 2\mathbb{m} = \{0, \pm 2, \pm 4, \pm 8, \dots\}$ is a subgroup.

$H_2 = 3\mathbb{m} = \{0, \pm 3, \pm 6, \pm 9, \dots\}$ is a subgroup.

$H_3 = 4\mathbb{m} = \{0, \pm 4, \pm 8, \pm 12, \dots\}$ is a subgroup.

Now $H_1 \cup H_2 = \{0, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 9, \dots\}$ is a subset of $\mathbb{Z}$

Here $2, 3 \in H_1 \cup H_2 \Rightarrow 2+3 \notin H_1 \cup H_2$.

$\therefore H_1 \cup H_2$ is a subgroup.

Also $H_1 \cup H_3 = \{0, \pm 2, \pm 4, \pm 6, \pm 8, \dots\} = H_1$, subgroup.

$\therefore H_1 \cup H_3$ is a subgroup.

Here $H_3 \subset H_1$.

Example: Let G be a group and $H = \{x \in G \mid xy = yx, \forall x \in G\}$ is a subset of G . Then show that H is a subgroup of G .

Sol:- Given $H = \{x \in G \mid xy = yx, \forall x \in G\}$

Let $x, e \in G$. Then $e \cdot x = x \cdot e = e$

So $e \in H$. $\therefore H$ is non-empty subset of G .

Let $a, b \in H$ and $x \in G$

$$\begin{aligned} \text{consider } (ab)x &= a(bx) = a(xb) = (ax)b \\ &= (xa)b \\ &= x(ab) \end{aligned}$$

$$\Rightarrow ab \in H$$

$$\therefore a, b \in H \Rightarrow a \cdot b \in H.$$

Also $a \in H, x \in G \Rightarrow xa = ax$

Since $a \in G \Rightarrow a^{-1} \in G$

pre operating and post operating with a^{-1} on both sides,

$$a^{-1}(xa)a^{-1} = a^{-1}(ax)a^{-1}$$

$$(a^{-1}x)(aa^{-1}) = (a^{-1}a)(xa^{-1})$$

$$(a^{-1}x)e = e(xa^{-1})$$

$$a^{-1}x = xa^{-1}$$

$$\Rightarrow a^{-1} \in H$$

$$\therefore a \in H \Rightarrow a^{-1} \in H.$$

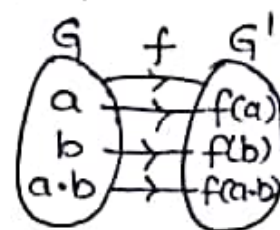
H is a subgroup of G .

Homomorphism: Let $(G, \cdot)$ and $(G', *)$ be

two groups. The mapping $f: G \rightarrow G'$ is

said to be a Homomorphism if

$$f(a \cdot b) = f(a) * f(b), \forall a, b \in G$$



i.e., Image of product = product of images.

If $f: G \rightarrow G'$ is a homomorphism then G' is called is called homomorphic image group. The groups G and G' are called homomorphic groups. It is denoted by $G \cong G'$.

One-one: A homomorphism $f: G \rightarrow G'$ is said to one-one, if different elements in G will have different images in G' .

$$\text{i.e., } a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$$

(or) If $f(a_1) = f(a_2)$ then $a_1 = a_2$

Range of f: In a homomorphism, the set of all image elements is called Range of homomorphism

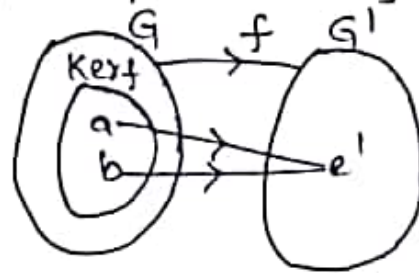
$$\text{Range } f = f(G) = \{ f(a) \mid a \in G \} \subset G'$$

onto: A homomorphism $f: G \rightarrow G'$ is said to be onto if range and codomain are equal.

$$f \text{ is onto} \Leftrightarrow \text{Range } f = \text{codomain.}$$

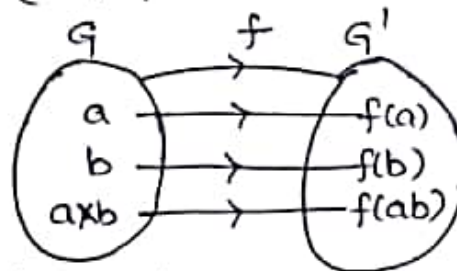
Isomorphism: A one-one and onto homomorphism is called Isomorphism. (X)

Kernel of Homomorphism: If $f: G \rightarrow G'$ is a homomorphism then Kernel of homomorphism is defined as $\text{Ker } f = \{a \in G \mid f(a) = e'\}$ where $e' \in G'$ is the identity.



Example (1): Let G be a group of positive real numbers under multiplication and G' is a group of all real numbers under addition. The mapping $f: G \rightarrow G'$ is defined by $f(a) = \log a, \forall a \in G$. Prove that $f: G \rightarrow G'$ is an isomorphism.

Sol: Given $G = (\mathbb{R}^+, \times)$ is a group and $G' = (\mathbb{R}, +)$ is a group.



Also given $f(a) = \log a, \forall a \in G$

To prove $f: G \rightarrow G'$ is isomorphism.

(i) To prove $f: G \rightarrow G'$ is Homomorphism:

Let $a, b \in G$ then $ab \in G$

Now $a \in G \Rightarrow f(a) = \log a$

$b \in G \Rightarrow f(b) = \log b$

$ab \in G \Rightarrow f(ab) = \log(ab) = \log a + \log b = f(a) + f(b).$

(ii) To prove f is one-one:

Let $f(a) = f(b), \forall a, b \in G$

$\Rightarrow \log a = \log b$

$\Rightarrow a = b$

$\Rightarrow f$ is one-one.

(iii) To f is on-to : Let $x \in G'$ (codomain)
Then there exists $a \in G$ such that $f(a) = x$.

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$$\Rightarrow \log a = x$$

$$\Rightarrow a = e^x > 0, \forall x \in \mathbb{R}$$

$$\Rightarrow a = e^x \text{ is a +ve real no.}$$

$\therefore f$ is onto.

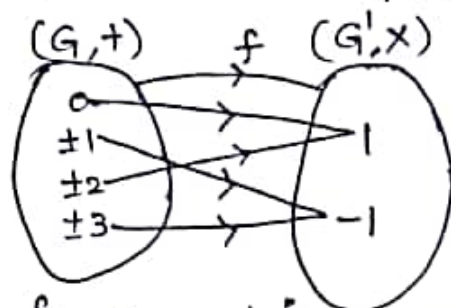
Hence $f: G \rightarrow G'$ is Isomorphism.

Example (2) : Let G be the additive group of integers and G' is the multiplicative group of $\{1, -1\}$. The mapping $f: G \rightarrow G'$ is defined as

$$f(a) = 1, \text{ if } a \text{ is even} \\ = -1, \text{ if } a \text{ is odd}$$

Then p.t f is homomorphism.

Sol :



To prove $f: G \rightarrow G'$ is a Homomorphism.

ie, we prove $f(a+b) = f(a) \times f(b), \forall a, b \in G$.

case (1) : Let 'a' and 'b' both are even.

Then $a+b$ is also even.

$$\therefore f(a) = 1, f(b) = 1 \text{ and } f(a+b) = 1$$

$$\therefore f(a) \times f(b) = 1 \times 1 = 1$$

$$\therefore f(a+b) = f(a) \times f(b).$$

case (2) : Let 'a' and 'b' both are odd. Then $a+b$ is even

$$\therefore f(a) = -1 \text{ and } f(b) = -1 \text{ and } f(a+b) = 1.$$

$$\therefore f(a) \times f(b) = -1 \times -1 = 1 = f(a+b)$$

$$\therefore f(a+b) = f(a) \times f(b)$$

case (3):- Let 'a' is even and 'b' is odd
Then $a+b$ is odd.

Now $f(a)=1$ and $f(b)=-1$

$$f(a+b)=-1$$

$$\therefore f(a) \times f(b) = 1 \times -1 = -1 = f(a+b)$$

$$\therefore f(a+b) = f(a) \times f(b)$$

$\therefore$ In all the cases, $f(a+b) = f(a) \times f(b)$

Hence $f: G \rightarrow G'$ is a homomorphism.